APL-II Nodejs

Node.js

- **Node.js** is a cross-platform, open-source server environment that can run on Windows, Linux, Unix, macOS, and more.
- Node.js is a back-end JavaScript runtime environment, runs on the V8 JavaScript Engine, and executes JavaScript code outside a web browser.
- Node.js has an event-driven architecture capable of asynchronous I/O
- Node.js was written initially by Ryan Dahl in 2009,
- The initial release supported only Linux and Mac OS X
- Node.js combined Google's V8 JavaScript engine, an event loop, and a low-level I/O API.
- asynchronous I/O (also non-sequential I/O) is a form of input/output processing that permits other processing to continue before the transmission has finished.

Node.js = Runtime Environment + JavaScript Library

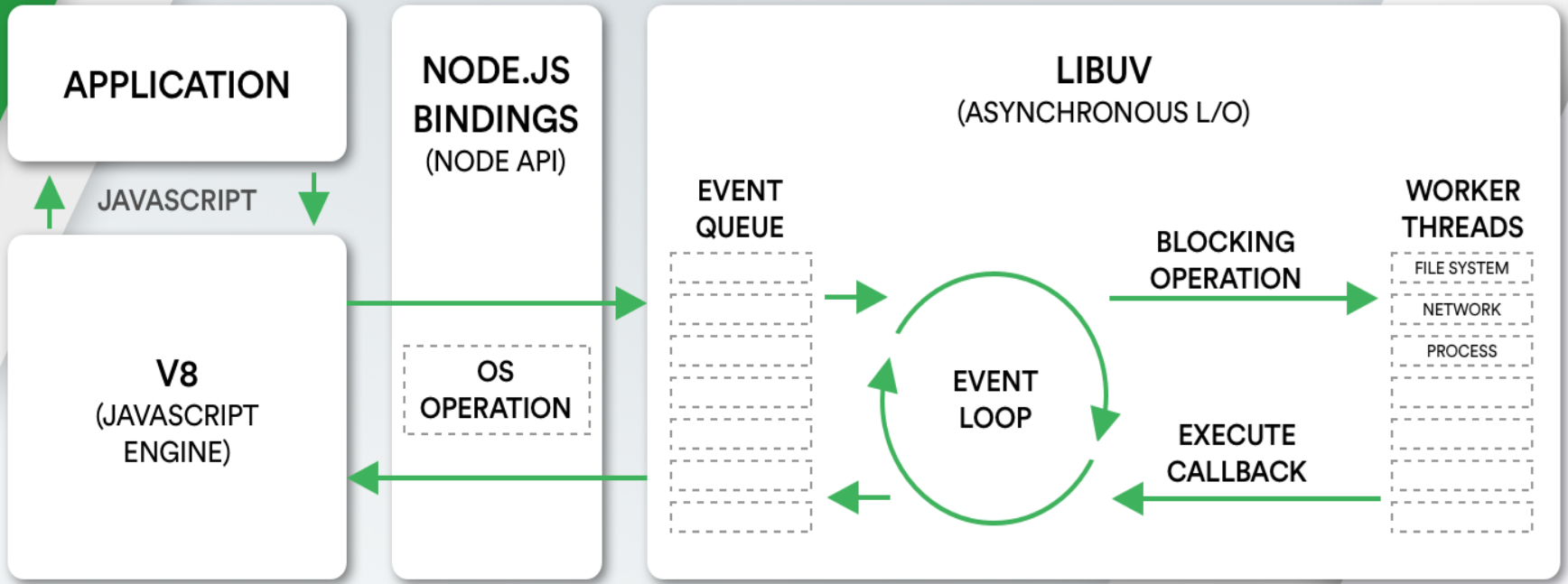
Node.js relies on various dependencies under the hood for providing various features.

- V8
- Libuv: is a C library
- Llhttp: new HTTP 1.1 parser
- C-ares: c-ares is a C library for asynchronous DNS requests (including name resolves)
- OpenSSL: c-ares is a C library for asynchronous DNS requests (including name resolves)

Concepts

- The cross-platform or multiplatform software is a type of application / program / software that works on various operating systems or devices, which are often called platforms. A platform means an operating system such as Windows, Mac OS, Android, or iOS
- In computer programming, a runtime system or runtime environment is a sub-system that exists both in the computer where a program is created, as well as in the computers where the program is intended to be run
- V8 is a free and open-source JavaScript and WebAssembly engine
- The code operates faster in server-to-client direction because of Javascript and Node.js.

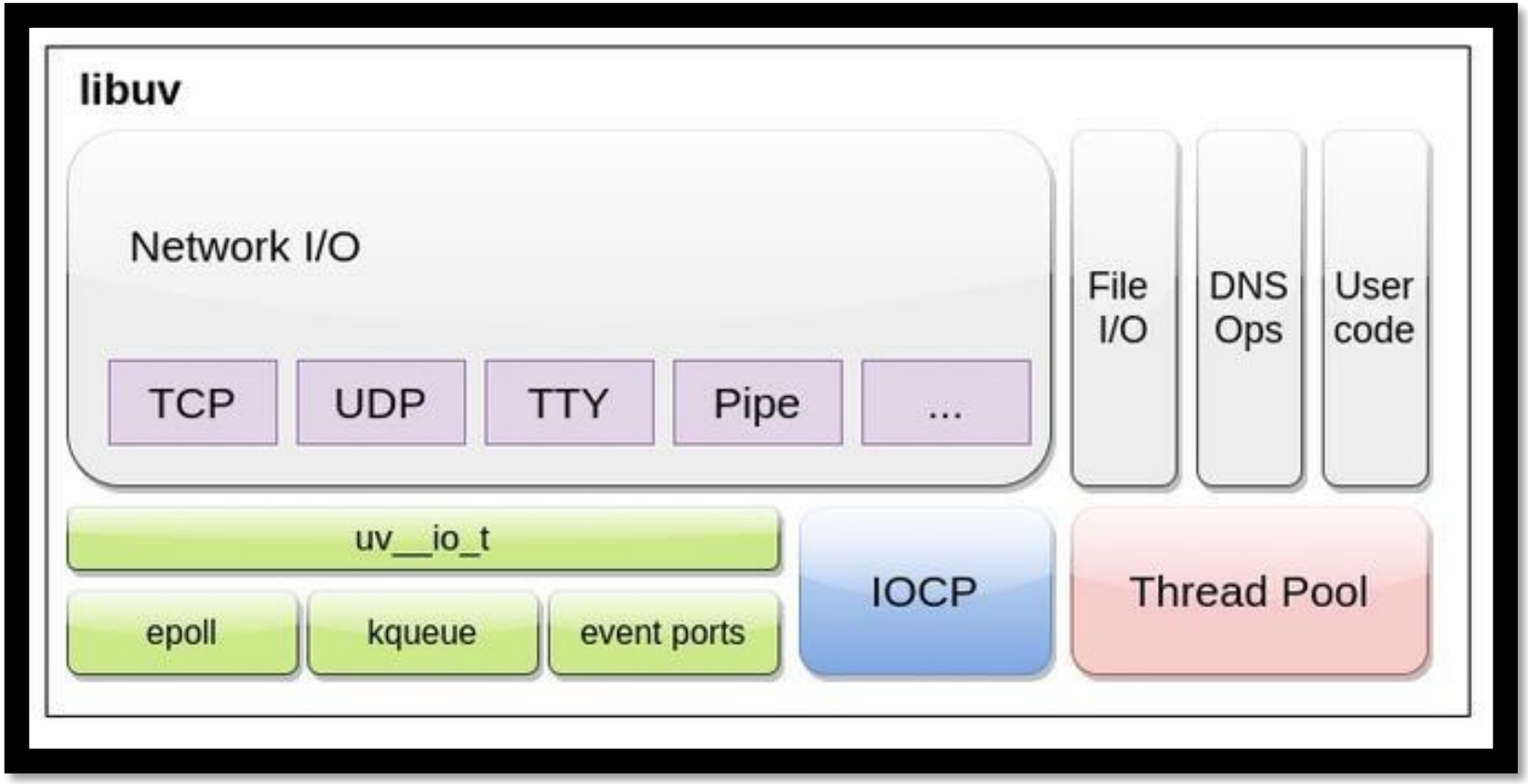
Node.js Architecture



Basics of Nodejs

- The Js website states that the language has 1,444,231 libraries and is still counting
- The best part is – web application development in Node.js enables a steady and secure non-blocking I/O model, simplifying the code
- Node.js runs on Google's V8 Javascript engine, where web applications are event-based in an asynchronous manner.
- Node.js platform uses a "*single-threaded event loop*."
- The platform does not follow a similar request/response multi-threaded stateless model; instead, it goes by a simplified single-threaded event loop model
- Node.js developers, a specific library called "*Libuv*" provides this mechanism known as an event loop.

- Node.js processing model is majorly based on Javascript event-based model along with the *callback mechanism*
- The traditional web-serving technique, where each request creates a new thread cramping up the system RAM
- Node.js operates on a single thread. This enables it to support thousands of concurrent connection handling event loops
- Libuv is a library written in the programming language C that helps nodejs to improve efficiency while running tasks parallelly.
- nodejs already have async API's.
- It uses Libuvs's thread pools if async API is not available in nodejs and processes are blocking the operations.


libuv
Network I/O
TCP
UDP
TTY
Pipe
...
File I/O
DNS Ops
User code
uv__io_t
epoll
kqueue
event ports
IOCP
Thread Pool

libuv:

libuv is a C library originally written for Node.js to abstract **non-blocking I/O operations**.

- Event-driven asynchronous I/O model is integrated.
- It allows the CPU and other resources to be used simultaneously while still performing I/O operations, thereby resulting in efficient use of resources and network.
- It facilitates an event-driven approach wherein I/O and other activities are performed using callback-based notifications.

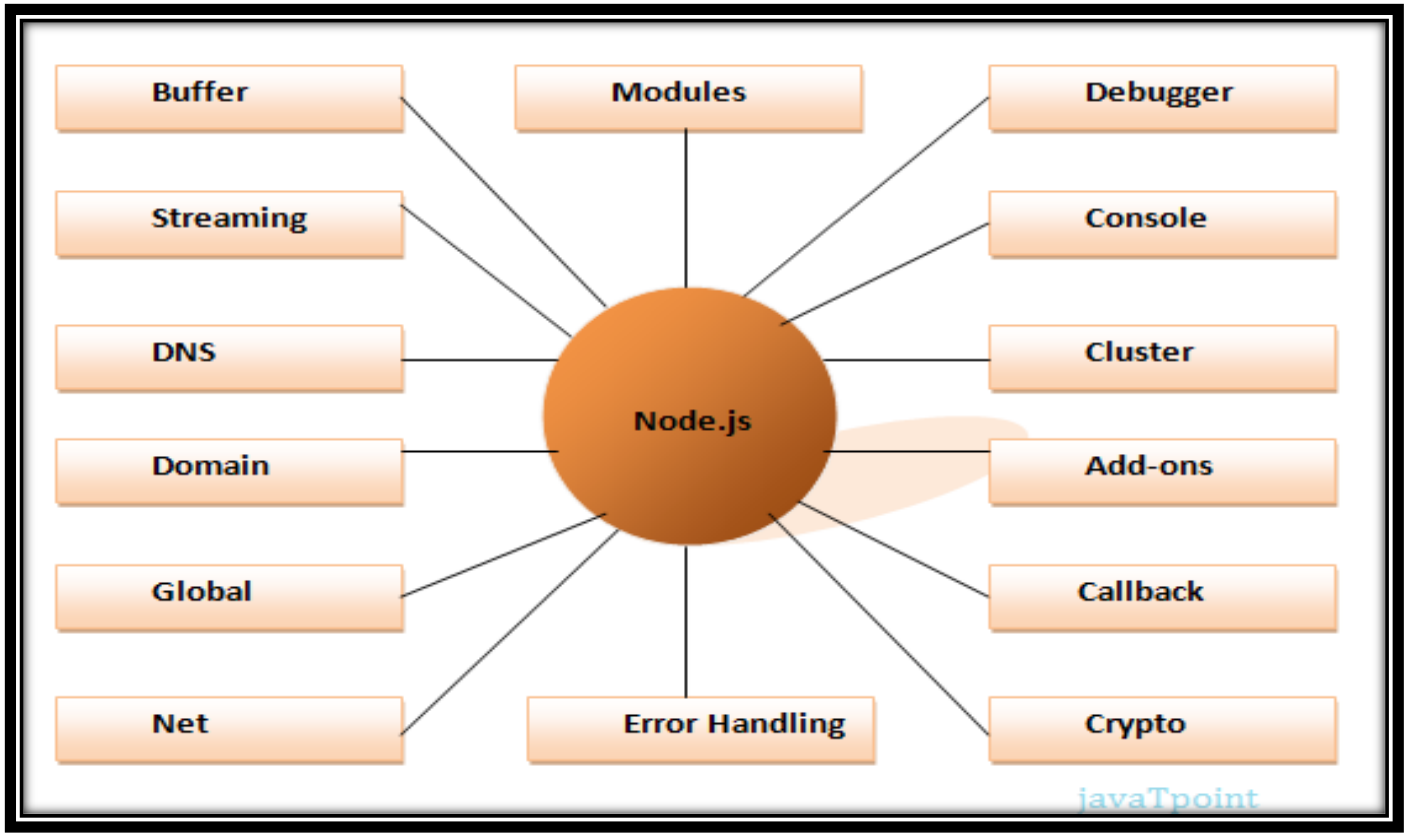
Key features

- Asynchronous/Non-blocking thread execution – Every API of the Node.js library is non-blocking. While waiting for a response for something outside the execution chain, the next tasks in the stack are continuously executed.
- Event-driven – A server built with Node.js uses a notification mechanism called “Events” to receive and track responses of previous API requests. Event Loop allows Node.js to execute all the non-blocking operations.
- Cross-platform compatibility – Node.js is compatible with various platforms like Windows, Linux, Mac OS X, Unix, and mobile platforms.

Event Loop

- the **event loop** is a programming construct or design pattern that waits for and dispatches events or messages in a program.
- The event loop works by making a request to some internal or external "event provider" (that generally blocks the request until an event has arrived), then calls the relevant event handler ("dispatches the event").
- The event loop is also sometimes referred to as the **message dispatcher**, **message loop**, **message pump**, or **run loop**.

Different parts of Node.js



Features of Node.js

- **Extremely fast:** Node.js is built on Google Chrome's V8 JavaScript Engine, so its library is very fast in code execution.
- **I/O is Asynchronous and Event Driven:** All APIs of Node.js library are asynchronous i.e. non-blocking. So a Node.js based server never waits for an API to return data. The server moves to the next API after calling it and a notification mechanism of Events of Node.js helps the server to get a response from the previous API call. It is also a reason that it is very fast.
- **Single threaded:** Node.js follows a single threaded model with event looping.

- **Highly Scalable:** Node.js is highly scalable because event mechanism helps the server to respond in a non-blocking way.
- **No buffering:** Node.js cuts down the overall processing time while uploading audio and video files. Node.js applications never buffer any data. These applications simply output the data in chunks.
- **Open source:** Node.js has an open source community which has produced many excellent modules to add additional capabilities to Node.js applications.
- **License:** Node.js is released under the MIT license.

Links

- https://www.google.com/search?sca_esv=553677874&cs=0&sxsrf=AB5stBhJ1wxcZq9bNSn5gvKhLmCRbmRquQ:1691140006831&q=Event+Loops+in+JavaScript&tbm=vid&source=lnms&sa=X&ved=2ahUKEwjhh_2608KAAxVISmwGHTR3AVAQ0pQJegQITRAB&biw=1920&bih=874&dpr=1#fpstate=ive&vld=cid:c1ddcda9,vid:vFJbKR6zfCE

- https://www.google.com/search?sca_esv=553677874&cs=0&sxsrf=AB5stBhJ1wxcZq9bNSn5gvKhLmCRbmRquQ:1691140006831&q=Event+Loops+in+JavaScript&tbm=vid&source=lnms&sa=X&ved=2ahUKEwjhh_2608KAAxVISmwGHTR3AVAQ0pQJegQITRAB&biw=1920&bih=874&dpr=1#fpstate=ive&vld=cid:aa83199a,vid:8aGhZQkoFbQ

Best

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